

Historical and future learning for the new era of multi-terawatt photovoltaics

Received: 2 June 2025

Accepted: 4 November 2025

Published online: 23 December 2025

 Check for updates

A list of authors and their affiliations appears at the end of the paper

Solar photovoltaics (PV) is entering a new era of multi-terawatt deployment, with 2 TW already in service and more than 75 TW predicted in many scenarios by 2050. This next era has been enabled by over five decades of cumulative advances in PV module cost reduction, performance and reliability. The current scale of deployment also introduces new needs, opportunities and challenges. In this Perspective we frame a path forwards based on learning, broadly defined as a combination of expansion of knowledge and advances through research and development, experience and collaboration. We discuss historical topics where learning has driven PV deployment until now, and emerging areas that are required to sustain high levels of future deployment. We expect progress to continue in terms of module price, performance and reliability, driven by advances in PV cell and module design, the emergence of tandem devices and increased focus on extending module lifetimes. Large-scale deployment also means large-scale sustainability and responsibility. We therefore posit that additional metrics, such as the impact on global CO₂ emissions, resource consumption and design for reuse and recycling, will become increasingly important to the PV industry and provide opportunities for further learning.

The global growth of photovoltaics (PV) for electricity generation has been remarkable, outpacing other energy technologies across history in rates of the expansion of manufacturing capacity and reduction in price^{1–3}. PV technology has been the largest source of new electricity installation worldwide for the past seven years and is starting to contribute an increasingly significant fraction of global electricity. Figure 1 documents key metrics for the rapid evolution of PV: cumulative deployment over time (Fig. 1a), cost reductions (Fig. 1b) and increases in efficiency (Fig. 1c) and reliability (Fig. 1d).

Participants at the 3rd Terawatt Workshop, one of a series of high-level international PV workshops led by Fraunhofer ISE, NREL and AIST and held in 2022, used target deployment values from literature studies for 100% renewable energy scenarios to sketch future expected installation. The result was a realistic trajectory to manufacturing levels of ~3 TW_{DC} (nameplate capacity denoted in direct current power) per year that would be needed to reach and sustain ~75 TW_{DC} of globally installed PV by 2050⁴. More recent studies have projected that additional, although largely linear, growth will be needed beyond that timeframe (Fig. 1a inset) if PV becomes a substantial primary

energy source^{5,6}. Expansion to that level (Fig. 1a) requires us to look ahead to new challenges and choices for PV to be made sustainable at scale. The rate and type of learning that has fuelled PV manufacturing and deployment to this point—a result of a global effort that has progressed over the decades—will continue to evolve. In the approach to multi-terawatt-scale annual manufacturing and deployment, PV has entered a new era.

Sustainability is at the heart of this new era. The module manufacturing capacity must increase to a level that meets (but does not outpace) the deployment demand. It is expected to become more incremental after the decades of exponential growth currently underway to drive the initial transition of electricity generation to PV (Fig. 1a). Correspondingly, the growth in PV manufacturing capacity is also anticipated to become more incremental as it nears ~3 TW yr^{−1} (ref. 4).

We must also ensure that the anticipated levels of future PV deployment can be supported by a global manufacturing infrastructure while also minimizing adverse societal and environmental impacts. Multiple factors have contributed to the price reduction that made PV a viable source of electricity, including cumulative experience, breakthroughs

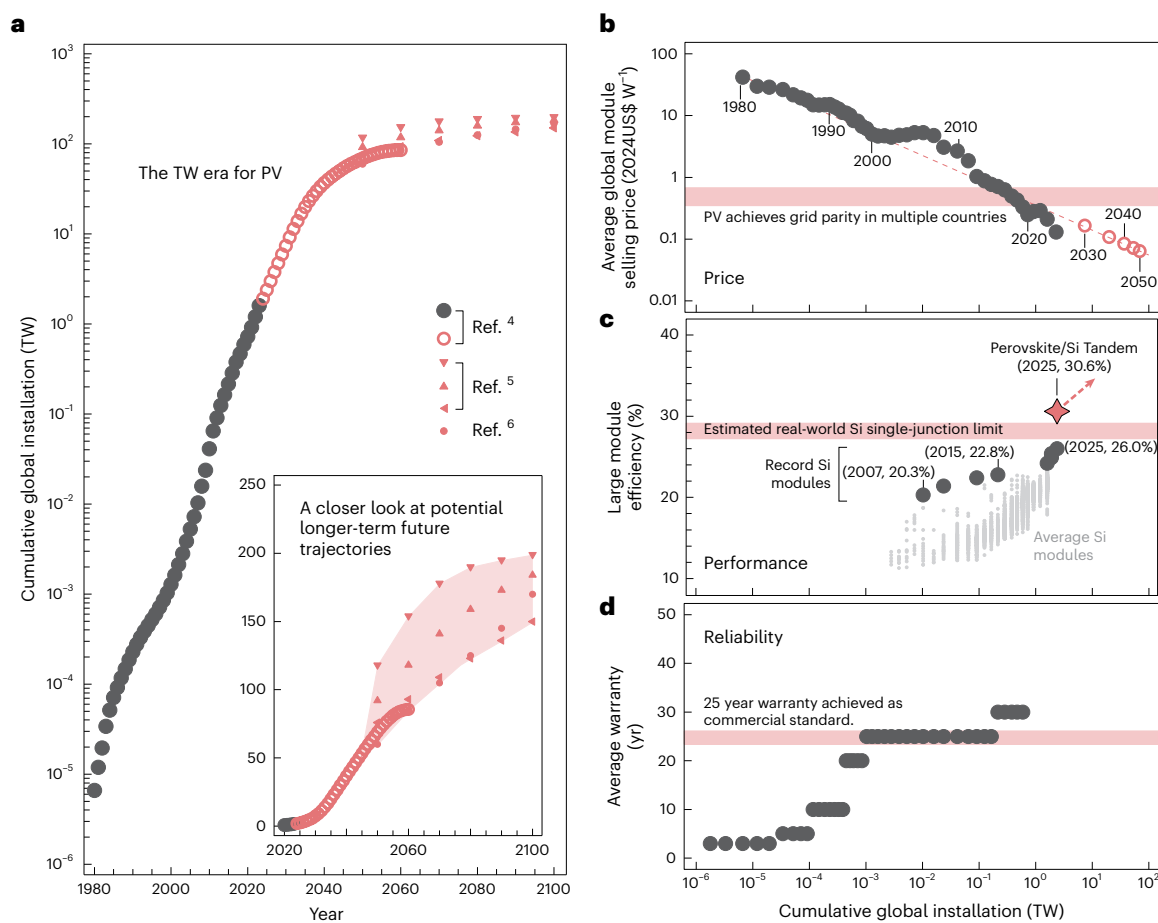


Fig. 1 | Topics of historical learning for PV development. a, Global cumulative PV installation as a function of time from 1980 to 2023 (filled grey circles) with an estimated trajectory (open red circles) to reach a global installation of $\sim 75 \text{ TW}_{\text{dc}}$ by 2050. The shaded region denotes a spread in projected trends in the literature. Data were obtained from refs. 4–6. The inset presents a more detailed view of future cumulative global installation scenarios up to 2100 on a linear scale. **b**, Global average selling prices of single-junction modules in 2024 inflation-adjusted prices (filled grey circles; data from ref. 12) and estimated future values (open red circles) based on extrapolation of the historical trend

(dashed red line). **c**, Efficiencies of commercial Si modules (small light-grey dots) obtained from an artificial intelligence algorithm that extracted relevant information from module datasheets for the period 2003–2023 for various module manufacturers worldwide (Methods). Also shown are record Si modules (filled grey circles; data from ref. 15) and tandem modules (red star; data from ref. 18), and a general direction of projected efficiency of tandem modules (red dashed arrow), all measured under one-Sun conditions. **d**, Average module warranty as a function of cumulative global installation (determined from data in Fig. 1a). Data from ref. 16.

delivered through research and development (R&D) and knowledge spillovers from other industries^{7,8}. This learning will continue, but additional areas of focus around sustainable design and manufacturing will grow in importance. These advances must be intentionally pursued, as meaningful progress is unlikely to emerge from cumulative experience alone.

In this Perspective we review recent progress in the development of PV technology and look ahead to identifying R&D activities that will be needed for this future. Cost reduction is expected to remain a critical driver of future progress, along with improvements in efficiency and reliability. A key component of future growth will also be ‘learning for sustainability’—that is, developing the knowledge, approaches and manufacturing ability to produce PV at multi-terawatt scale within the finite constraints of the Earth’s resources. We note at the outset that our assessment focuses on PV technology (that is, the module itself) and does not extend to other important aspects for achieving high levels of PV deployment, such as system design and pairing with storage to reduce generation variability.

Historical ‘learning’ in PV

One way improvements have historically been examined is through the framework of the experience curve. The experience curve is an empirical mathematical concept that aims to capture how an industry

improves proficiency at a particular task as a function of cumulative experience. It is an effective analytical tool in many domains, including the cost of manufacturing, labour productivity and the reduction of material use^{9–11}. In the PV domain, it is often presented as price versus cumulative installations or time rather than manufacturing cost due to availability of historical data. We present the curve for single-junction PV modules in price (Fig. 1b), where reductions have been driven largely by silicon (Si) modules¹².

The experience curve may be applied as long as the metric being analysed is improved by experience. The experience curve is not simply a model of economies of scale, although production volume learning is captured. In addition, the learning rate can evolve over time and is accelerated by sharing experience, particularly across the globe. Notably, clusters of PV companies, first in the United States, then in Japan and Germany and today in China, have shared knowledge through the training of new employees, active collaboration and the cross-licensing of intellectual property. Regularization of wafer and module size has helped production tool manufacturers to lower development costs, and the implementation of software tools for simulating solar cells, modules and systems has accelerated PV system design. Broader adoption of smaller efforts has also influenced how the PV community works together, for example, US government support of the Jet Propulsion

Laboratory's Block V reliability improvement program that shaped the current IEC 61215 qualification testing sequences¹³.

R&D has also been central to advances captured in the experience curve, yet it is a separate factor from experience⁸. Notably, directed R&D is an additional and valuable form of learning that sets the foundation for improvements. New cell and module designs that lead to cost reductions are one example. As the needs of the industry shift, emphasis on where to focus R&D will follow suit. In this sense, we use the term 'learning' throughout this Perspective as a proxy for the complex set of factors, including R&D, that contribute to improvements across many metrics.

Two other key performance parameters have also benefited from PV R&D coupled with experience: module efficiency and reliability. Module efficiency (Fig. 1c) is a key indicator of technical progress and benefits disproportionately from R&D breakthroughs. Its increase has hastened the economic viability of PV through decreases in the module area and corresponding balance of systems. Figure 1c also shows how record Si modules now approach the theoretical limit for Si single-junction technology^{14,15}.

Likewise, module reliability has increased through the study of degradation mechanisms and continuous feedback from field performance. One indirect but readily available measure of reliability that reflects market expectations is the average number of years of module warranty (Fig. 1d)^{16,17}. We plot warranty as a function of cumulative global installation to indicate the role of experience, consistent with the common presentation of price reduction.

The future of learning in PV

Drawing from this historical backdrop, we look towards future R&D needs in this new era for solar PV. We expect a reliance on continued progress in price, performance and reliability to drive PV installations. We also anticipate that additional factors will emerge to reach manufacturing and deployment goals with available resources. This includes resource and material availability. We predict that net emissions, embodied energy, resource consumption and recycling will increasingly be considered in new designs and manufacturing approaches.

Progress in historical learning for the near term

The future trajectory of the price/cost experience curve for PV modules is an interesting topic for debate. Even with continued manufacturing growth, the increments of cost decrease become smaller for every magnitude of additional deployment. As industries mature, cost minimization strategies are often increasingly coupled with or replaced by performance-maximizing strategies. Here we address the question of what advances in the near and longer terms will enable continued progress in PV performance and cost. Innovations in wafer production, cell and module design, manufacturing steps and balance-of-system components are traditional areas of focus for the PV industry. Improvements in module efficiencies, reliability, installation practices (for example, automation), engineering, permitting requirements and operations protocols are other cost and system performance drivers. With these advances, the PV industry has a demonstrated track record of surpassing earlier estimated 'limits' in module cost, performance and broader integration. We believe that there is more room to run as PV use expands worldwide.

Emerging technologies, such as thin-film metal halide perovskites and tandem architectures, have also gained interest as potential enablers of future manufacturing-related cost reduction and efficiency gains, respectively. Several tandem architectures currently considered incorporate thin-film junctions into their designs to achieve both objectives. Substantial discussion has been devoted to the topic of tandem architectures as the next major step in the evolution of PV, addressing questions such as "what role will tandems play in the global distribution of PV installed by 2050 and beyond?" and "what learning is needed for tandems to be widely deployed?".

The main advantage of tandem architectures is their potential boost in efficiency. Whereas the record Si module efficiency stands at 26.0%, module efficiencies are expected to peak at around 27% (refs. 14,18). Tandem technologies can surpass this limit, with small-area perovskite/Si cells and modules now demonstrated at 34.85 and 30.6%, respectively¹⁸. With sizeable investment, Si-based tandem modules may be expected to realize ~0.5% annual efficiency gains up to 2030, and are eventually predicted to reach >35% in the next decades^{14,19}. Higher efficiencies help to reduce the balance of systems and module cost per watt, if the relative watts gained exceed the relative increase in total cost and the degradation rates and warranties are similar to Si modules today. All-thin-film tandem architectures may enable manufacturing with an even lower cost per watt, raw material consumption, capital expenditure and carbon intensity. Finally, innovations in tandem architectures and manufacturing may allow new industry players to enter the market, leading to a more globally diversified cell- and module-manufacturing landscape and supply chain.

Despite their long-term efficiency potential, tandem solar cells (and in particular perovskite junctions) still require substantial improvements in reliability and scalability before they are ready for mass deployment. Manufacturing processes are also still being worked out. Tandem solar cells will undergo experience-based learning in all three areas, aided by previous lessons in Si, cadmium telluride (CdTe), CIGS (that is, copper indium gallium selenide) and III–V modules. Those technologies demonstrated the importance of R&D in addition to learning through cumulative production in unlocking their potential. Early adoption markets will also serve as valuable proving grounds. Questions remain about whether and when tandems will be competitive for mass deployment on module cost alone, or whether they can progress along the associated experience curve rapidly enough to meaningfully contribute to the buildout of manufacturing capacity for 3 TW yr⁻¹ by 2050.

A consensus view from the authors of this Perspective is that tandem architectures will be developed and deployed, given the balance-of-systems multiplier benefits associated with increased efficiency. Tandem architectures have the added advantage that they can sometimes leverage existing manufacturing infrastructure (particularly in the case of perovskite/Si tandems) to speed manufacturing scale-up. Collaboration across the global PV R&D community and a ramp-up in manufacturing are important for accelerating the pace of these advances. An increasing emphasis on new sustainability metrics may also shift the balance to those PV technologies that are best able to support them.

Reliability plays a critical role in the economics and bankability of PV installations and the ability to meet global demand (that is, keeping modules in the field for longer reduces the need to install new ones). As with the case of cost, we may also ask "what is reliable enough?", particularly in the case of next-generation, very-high-efficiency modules. **A prevailing view is that new modules will have to be at least as reliable as established ones to gain widespread adoption in terawatt-scale markets** (although niche markets may be more accepting of higher degradation/failure rates). New technologies must progress through four steps: (1) specification of the module performance for controlled operating conditions, (2) reaching a predictable short-term fielded energy output, (3) identification of the potential for early failure through standardized accelerated stress tests and (4) the mitigation of risks for new/unknown performance degradation modes. The fourth item remains an issue even for today's Si technologies, as product evolution (for example, larger modules, thinner glass or evolving device designs) introduces new risk.

Emerging perovskite and tandem technologies require attention in all four measures. Extensive and transparent investigation to identify and understand degradation and failure modes, establish accelerated stress tests to probe those modes, and create strategies to mitigate them will help to make these technologies viable at scale. This work cannot be done by either academia or industry alone. Technical

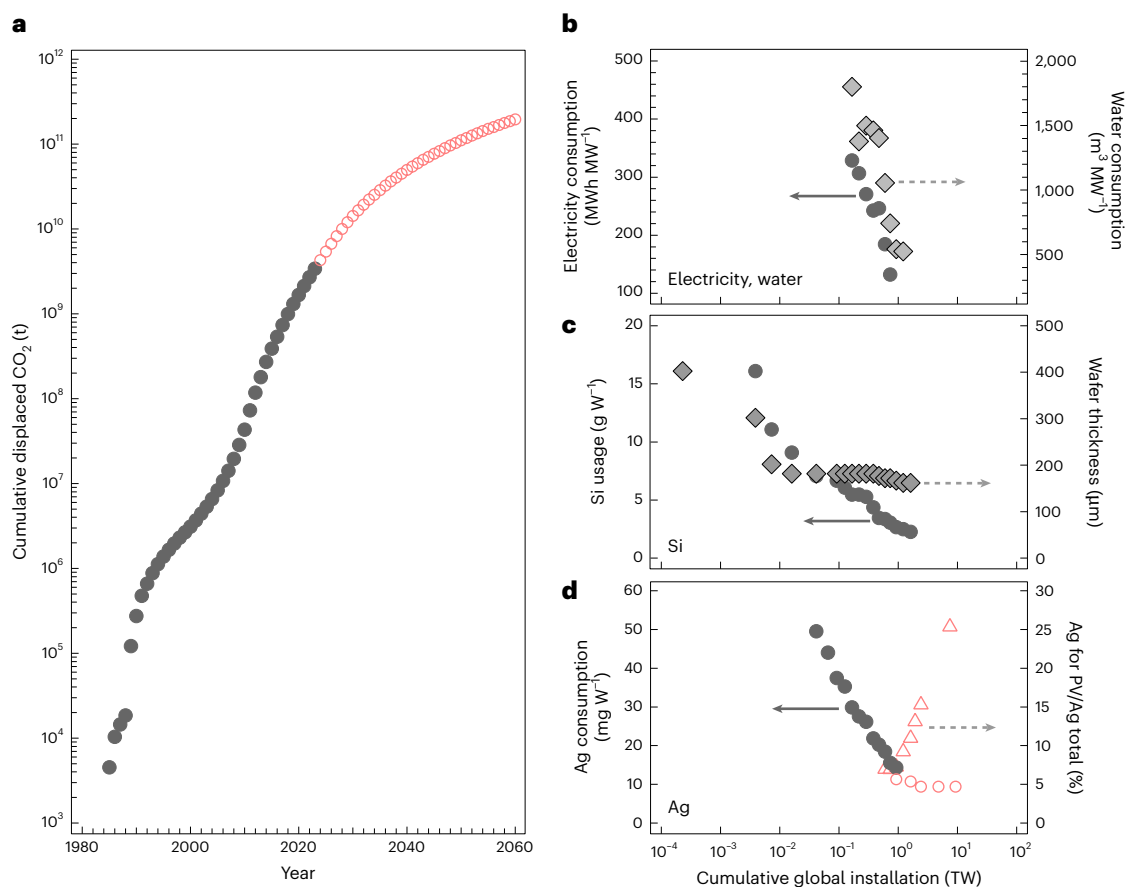


Fig. 2 | Future of PV learning for sustainability. **a**, Global cumulative displaced tons of CO_2 as a function of time from 1980 to 2023 (filled grey circles) with a projected trajectory (open red circles) associated with a cumulative installation of ~ 75 TW of PV by 2050. The data were obtained from the approach noted in the Methods. **b**, Consumption of manufacturing electricity (left axis, filled grey circles) and water (right axis, filled grey diamonds) averaged over three large Si and CdTe manufacturers. These values only capture part of the cell- and

module-manufacturing process and do not account for other sources of energy, such as heat. Data were obtained from refs. 26–28. **c**, Average Si usage (left axis, filled grey circles) and Si wafer thickness (right axis, filled grey diamonds). Data were sourced from past International Technology Roadmap for Photovoltaics reports²⁹. **d**, Ag usage (left axis, filled grey circles for actual and open red circles for projected consumption) and projected fraction of global Ag use (open red triangles) as a function of cumulative global installation (determined from data in Fig. 1a).

specifications and standards developed by the entire community, centralized and accessible facilities for long-term reliability testing, and beachhead markets for initial smaller-scale deployment will also help to accelerate progress.

New learning for the next era of PV

What additional areas of focus for R&D, manufacturing innovation and collaboration are needed to sustain terawatt-scale PV? Energy, water and material-reduction strategies, combined with circular manufacturing approaches to further minimize the energy, carbon and waste footprints of PV, are all important aspects to consider. Energy sustainability at the system level, measured as an energy return on investment metric, in turn benefits from the underlying electricity produced by PV technologies^{20–22}. R&D focus on these aspects now will contribute to the long-term success of PV as a major energy source and a sustainable, profitable industry.

Figure 2 presents important sustainability-related metrics, using the experience curve format, analogous to Fig. 1. One primary benefit of PV is reduced greenhouse gas (GHG) emissions for electricity generation²³. The amount of carbon dioxide (CO_2) emissions that PV displaces as it replaces other forms of electricity generation^{4,24} (Fig. 2a) is expected to grow with cumulative installation. For context, the total amount of CO_2 emitted by humans in the past 275 years is estimated at 1,800 gigatons (ref. 25).

The consumption of energy and water per module are important metrics for minimizing the environmental impact. Figure 2b shows

averages of these values reported by three companies—two Si module manufacturers (Trina and Jinko) and a CdTe module manufacturer (First Solar)—as examples^{26–28}. Both have decreased over time. Past gains have been made as manufacturers have increased wafer and module sizes and the throughput of manufacturing tools. Going forwards, one route to reducing embedded carbon in the module is to reduce the intensity of its manufacturing energy. This approach may become more accessible as new manufacturing processes and module designs are implemented with energy intensity in mind, and scaling continues for thin-film and/or tandem PV technologies.

Improving material utilization is a parallel route for lowering GHG emissions, the embodied energy and cost. Silicon consumption per watt has come down eightfold in the past two decades (Fig. 2c), driven by reductions in wafer thickness and kerf loss, new methods for utilizing a greater fraction of the Si boule and increases in cell and module efficiencies^{12,29}. R&D focused on removing steps in the Si wafer production process promises to yield additional gains. Silver (Ag) consumption (Fig. 2d) is critical from the standpoint of overall availability^{30,31}. Usage per watt of panel power generation has greatly decreased over the past two decades, but the rate of future Ag reductions is expected to slow. At the same time, anticipated future growth of Si module production will lead to an increase in overall consumption and the percentage of Ag supply utilized by the PV industry. Current R&D is focused on developing processing methods that use less Ag or replace it with other metals^{32,33}. For example, copper plating and

Table 1 | Summary of areas for future R&D into PV

Areas for R&D from the 4th Terawatt Workshop	
Cost	Continue innovating in manufacturing, performance and the reduction of resource consumption
	Replace scarce materials that will not be able to reach multi-terawatt production levels
	Research existing and next-generation thin-film PV technologies
	Standardize wafer and module sizes and supply chains
	Tandems: reduce capital expenditure for manufacturing
	Module assembly: pilot lines for module innovation
Performance	Continue the development of advanced tunnel oxide passivated contact (TOPCon) and back-contact (XBC) solar cell designs
	Reduce the long-term degradation rates in high-efficiency cells
	Understand metastable defects in high-efficiency solar cells, and avoid or passivate them
	Tandems: support multiple paths towards commercial tandem PV modules
	Module assembly: improve light trapping, upshift infrared photons to increase potential maximum theoretical efficiency limits
Reliability	Design PV modules and systems for longer lifetimes
	Reduce degradation rates (long-term goal: $<0.1\% \text{yr}^{-1}$)
	Tandems: replicate the reliability framework already in place for established PV technologies
	Advance the science and application of combined accelerated stress testing
Resource consumption	Increase the mix of clean energy used to produce modules
	Reduce energy, water, material consumption and CO ₂ emission
Material use	Innovate methods for recycling and incorporating recycled materials into modules
	Develop methods to remove impurities from recycled materials
	Use raw materials that are locally and sustainably sourced
	Improve the material usage efficiency while maintaining the performance
Design for sustainability	Design modules for disassembly and recycling
	Establish large-scale markets for circular approaches (that is, horizontal recycling rather than downcycling and so on)
Design for adoption	Design for local constraints and to drive adoption regionally
	Expand the design and implementation of distributed PV for global development
	Enable a future of 'PV everywhere' by designing for dual use and system integration

Topics span the historical (cost, performance, reliability) and emerging (resource consumption, material use, design for sustainability) categories in addition to an overarching need in the area of design for adoption. Throughout, global learning via the exchange of people and information is critical to achieving the learning objectives.

screen-printing methods are being developed for a range of cell technologies, whereas pattern transfer printing for metallization is being implemented to reduce Ag finger widths³⁴.

Glass usage has likewise been reduced over the years³⁵. Whereas material reduction is generally positive from the standpoints of GHG emissions and energy consumption, it introduces design trade-offs. The recent move to larger Si modules coupled with the use of thinner glass coversheets (as low as 0.5 mm) has led to an increase in the mechanical failure rate of modules^{35,36}. Materials thrifting must be supported with reliability research that is specific to climate zones, applications and new module designs^{37,38}. Trends towards more widespread use of glass–glass module designs may also impact the trajectory of total glass usage in the future.

Design for reliability, reuse and recycling represents a third route to enhanced sustainability³⁹. Increasing module lifetime is one route towards cost effectiveness and reduced emissions, as fewer modules must be manufactured as replacements over time. This is yet another benefit of learning for reliability. Designing for repair and reuse may have similar benefits, but this must not impair the module life or economic viability. Recycling is receiving increasing attention and can potentially become part of the supply chain⁴⁰. Although current volumes of decommissioned PV modules are not yet large enough, now is the time for R&D on module design for disassembly and recycling technologies. As the volume of decommissioned PV modules increases, materials science R&D, enhanced recycling process efficiencies, specialization for local markets and improved logistics may also drive

down recycling costs and increase the amount of recovered material. Specific challenges include preventing the introduction of impurities into the recovered material, navigating an increasing bill of materials across a range of PV technologies and recovering high-value materials such as Si, metals and glass.

These and other sustainability-focused metrics that emerge in the coming years will be important for the PV community to address head-on, as they are unlikely to be advanced through cumulative experience alone. We predict that focused R&D efforts and collaboration will be concentrated in these areas going forwards. The improvements they yield will enable the manufacturing capacity and deployment of PV to continue to be scaled to the levels needed to meet new electricity demand.

Outlook

Table 1 summarizes additional topics for learning in the relevant historical (cost, efficiency and reliability) and emerging (resource consumption, materials use and design for sustainability) categories. Learning for sustainability has already helped to drive down costs and will be even more critical for the PV industry to reliably access the resources needed for future deployment demands.

However, some essential R&D activities may not yield immediate financial incentives for industry to take on alone. The entire PV community must address these needs through continued R&D investment, shared experience, global collaboration and cooperation, standardization and a healthy educational environment. New metrics are also

expected to emerge as PV fully transitions from an upstart technology to a major global energy source. Global redistribution of PV module manufacturing could further help to reduce supply-chain bottlenecks, improve access to PV and enhance its acceptance across more communities that directly benefit from its production and installation.

The pace of learning in these categories will necessarily evolve throughout the transition to the multi-terawatt era of PV. Global installations are proceeding along the trajectory previously projected by this group, in large part due to the investments made by the PV industry⁴. Maintaining this momentum will go a long way to reaching the 2050 goals. New goals set by the PV community will carry it into the future as a majority of its deployment objectives are met and manufacturing growth is no longer the main driver for industry. This mindset will be new for a community that has until now focused largely on growth and cost reduction.

We must also not lose sight of the reason for advancing PV deployment in the first place: securing an abundant, affordable, resilient and sustainable source of energy for humanity. Learning will also continue in the design of PV modules and systems to serve a wider range of communities, geographic and economic constraints and applications. Topics for future PV community meetings, such as the 4th Terawatt Workshop that informed this Perspective, may shift to addressing system and end-user needs. Investment, manufacturing and adoption today will pay globally transformative dividends tomorrow in terms of economic growth, productivity, job creation and reduced pollution and poverty. A focus on how the global PV community learns together will unlock this promising future.

Methods

Average Si module data

For the average Si module data presented in Fig. 1c, artificial intelligence was used to detect and classify tables in module datasheets that were provided as PDF (portable document format) files. The in-house developed tool 'Lightning-table'^{41,42} for the end-to-end pipeline is divided into three major parts: (1) table-containing areas of the datasheets are detected using deep-learning-based object detectors, (2) the raw values presented in the tables are extracted using off-the-shelf table extraction utilities and (3) the raw values are structured and validated using a combination of machine learning and rule-based methods. For (1), the library PyMuPDF⁴³ was used to convert the pages of the datasheets into images and the deep learning library PyTorch⁴⁴ together with object detector Faster R-CNN v2⁴⁵ was used to detect the tables in the images. For (2), the utilities Tabula⁴⁶ and Camelot^{47,48} were used together with optical character recognition⁴⁹ to extract the data in the tables. For (3), the word vectorization technique known as term frequency-inverse document frequency (or TF-IDF)⁵⁰ and the naive Bayes⁵¹ classifier were implemented in Scikit-learn⁵² for structuring of the raw values. The complete pipeline achieved 99.9% precision and a 94.9% recall performance at extracting electrical, thermal and mechanical characteristics of tables in the datasheets from 2003 to 2023.

The dataset for training and evaluation was compiled from datasheets of PV modules obtained directly from manufacturers, either via their official websites or scanned from original printed materials. These datasheets span a period of approximately two decades. Data were sampled randomly from a larger pool of files, including 420 module datasheets from 140 manufacturers, resulting in a training dataset comprising 2,675 images. The total number of tables was 5,896 because most of the images contained multiple tables. Hyperparameters of the network were trained for 50 epochs, lasting about eight hours, with a batch size of four and a learning rate of 10^{-4} . Eighty per cent of the dataset was used for training and 20% of the dataset was used for evaluation. Owing to the age of many documents, the majority of the original web links are no longer active, and not all datasheets remain publicly accessible. Consequently, whereas some of the data may still

be found online, the full set of 420 datasheets from 140 manufacturers is no longer entirely available in the public domain.

Calculation of cumulative displaced CO₂ data

The values for Fig. 2a were calculated by multiplying the relative contribution of PV energy to electricity generation G_{PV} (given in per cent) with total global electricity generation E_{tot} (given in terawatt hours), and multiplying that with the global average carbon intensity CI_{tot} (given in grams per kilowatt hour). The resulting estimate is conservative because industrially highly developed countries with high PV installation rates tend to have higher than average carbon emissions. Embedded carbon was estimated but only results in a minor correction (<3%) of the obtained value. Data for this calculation were taken from the following datasets: electricity generation by source from ref. 53 and carbon intensity from ref. 24.

Electricity and water consumption data were obtained from published sustainability reports from the companies Trina, Jinko and First Solar. Because the total scope of the manufacturing processes included in these values may vary between reports from different companies, it is not advisable to directly compare values between them. The data were therefore averaged over all three companies.

References

1. *The Energy Security Scenarios: Full Report 18* (Shell, 2023); <https://www.shell.com/news-and-insights/scenarios/the-energy-security-scenarios.html>
2. Way, R., Ives, M. C., Mealy, P. & Farmer, J. D. Empirically grounded technology forecasts and the energy transition. *Joule* **6**, 2057–2082 (2022).
3. Lempriere, M. Wind and solar are growing faster than any other sources of electricity in history, according to new analysis from thinktank Ember. *Carbon Brief* <https://www.carbonbrief.org/wind-and-solar-are-fastest-growing-electricity-sources-in-history/> (2024).
4. Haegel, N. M. et al. Photovoltaics at multi-terawatt scale: waiting is not an option. *Science* **380**, 39–42 (2023).
5. Keiner, D., Gulagi, A. & Breyer, C. Energy demand estimation using a pre-processing macro-economic modelling tool for 21st century transition analyses. *Energy* **272**, 127199 (2023).
6. Goldschmidt, J. C., Wagner, L., Pietzcker, R. & Friedrich, L. Technological learning for resource efficient terawatt scale photovoltaics. *Energy Environ. Sci.* **14**, 5147–5160 (2021).
7. Kavlak, G., McNerney, J. & Trancik, J. E. Evaluating the causes of cost reduction in photovoltaic modules. *Energy Policy* **123**, 700–710 (2018).
8. Nemet, G. F. Beyond the learning curve: factors influencing cost reduction in photovoltaics. *Energy Policy* **34**, 3218–3232 (2006).
9. Wolf, M. Historical development of solar cells. In *Proc. 25th Power Sources Symposium* 120–124 (PSC Publications Committee, 1972).
10. Sopori, B., Sinton, R., Tan, T. & Swanson D. *13th Workshop on Crystalline Silicon Solar Cell Materials and Processes: Summary Discussion Report* NREL/CP-520-35435 (National Renewable Energy Laboratory (NREL), 2004); <https://www.nrel.gov/docs/fy04osti/35435.pdf>
11. Swanson, R. M. A vision for crystalline silicon photovoltaics. *Prog. Photovolt.* **14**, 443–453 (2006).
12. Philipps, S. & Warmuth, W. *Photovoltaics Report* (Fraunhofer ISE, 2024); <https://www.ise.fraunhofer.de/en/publications/studies/photovoltaics-report.html>
13. Kurtz, S. et al. *Photovoltaic Module Qualification Plus Testing* Technical Report NREL/TP-5200-60950 (NREL, 2013); <https://doi.org/10.2172/1118100>.
14. Hermle, M., Feldmann, F., Bivour, M., Goldschmidt, J. C. & Glunz, S. W. Passivating contacts and tandem concepts: approaches for the highest silicon-based solar cell efficiencies. *Appl. Phys. Rev.* **7**, 021305 (2020).

15. Champion photovoltaic module efficiency chart. NREL <https://www2.nrel.gov/pv/module-efficiency>
16. Jordan, D. C., Haegel, N. & Barnes, T. M. Photovoltaics module reliability for the terawatt age. *Prog. Energy* **4**, 022002 (2022).
17. Peters, I. M., Hauch, J., Brabec, C. & Sinha, P. The value of stability in photovoltaics. *Joule* **5**, 3137–3153 (2021).
18. Green, M. A. et al. Solar cell efficiency tables (version 66). *Prog. Photovolt.* **33**, 795–810 (2025).
19. Cordell, J. J., Woodhouse, M. & Warren, E. L. Technoeconomic analysis of perovskite/Si tandem solar modules. *Joule* **9**, 101781 (2025).
20. Görig, M. & Breyer, C. Energy learning curves of PV systems. *Environ. Prog. Sustain. Energy* **35**, 914–923 (2016).
21. Louwen, A., van Sark, W. G. J. H. M., Faaij, A. P. C. & Schropp, R. E. I. Re-assessment of net energy production and greenhouse gas emissions avoidance after 40 years of photovoltaics development. *Nat. Commun.* **7**, 13728 (2016).
22. Sahin, H., Solomon, A. A., Aghahosseini, A. & Breyer, C. Systemwide energy return on investment in a sustainable transition towards net zero power systems. *Nat. Commun.* **15**, 208 (2024).
23. Wikoff, H. M., Reese, S. B. & Reese, M. O. Embodied energy and carbon from the manufacture of cadmium telluride and silicon photovoltaics. *Joule* **6**, 1710–1725 (2022).
24. Carbon intensity of electricity generation, 2024. *Our World in Data* <https://ourworldindata.org/grapher/carbon-intensity-electricity> (2024).
25. Annual CO₂ emissions. *Our World in Data* <https://archive.ourworld-indata.org/20250624-125417/grapher/annual-co2-emissions-per-country.html> (2024).
26. Sustainability Reports 2018–2024 (First Solar, 2018–2024); <https://www.firstsolar.com/en/Resources/Sustainability-Documents>
27. Environmental, Social and Governance (ESG) and Corporate Social Responsibility (CSR) Reports (Jinko Solar, 2023); <https://jinkosolar.eu/downloads/sustainability-downloads/>
28. Sustainability and Corporate Social Responsibility Reports (Trina Solar, 2012–2023); <https://static.trinasolar.com/us/our-company/sustainability-downloads>
29. International Technology Roadmap for Photovoltaics (ITPRV) Reports (VDMA, 2025); <https://www.vdma.eu/international-technology-roadmap-photovoltaic>
30. Hallam, B. et al. The silver learning curve for photovoltaics and projected silver demand for net-zero emissions by 2050. *Prog. Photovolt.* **31**, 598–606 (2023).
31. Feltrin, A. & Freundlich, A. Material considerations for terawatt level deployment of photovoltaics. *Renew. Energy* **33**, 180–185 (2008).
32. Martinez-Szewczyk, M. W., DiGregorio, S. J., Hildreth, O. & Bertoni, M. I. Reactive silver inks: a path to solar cells with 82% less silver. *Energy Environ. Sci.* **17**, 3218–3227 (2024).
33. Zhang, Y. et al. Ultra-lean silver screen-printing for sustainable terawatt-scale photovoltaics. *Sol. RRL* **8**, 2400478 (2024).
34. Lossen, J., Matusovsky, M., Noy, A., Maier, Ch. & Bähr, M. Pattern Transfer Printing (PTP™) for c-Si solar cell metallization. *Energy Procedia* **67**, 156–162 (2015).
35. Gupta, M., Advancements in ultra-thin solar glass: benefits and challenges for modern photovoltaic systems. *SolarQuarter* <https://solarquarter.com/2024/07/26/advancements-in-ultra-thin-solar-glass-benefits-and-challenges-for-modern-photovoltaic-systems/> (2024).
36. Silverman, T. J. et al. *Tough Break: Many Factors Make Glass Breakage More Likely* Technical Report NREL/TP-5K00-91695 (NREL, 2024); <https://www.nrel.gov/docs/fy25osti/91695.pdf>
37. Ascencio-Vasquez, J., Brecl, K. & Topič, M. Methodology of Köppen-Geiger-Photovoltaic climate classification and implications to worldwide mapping of PV system performance. *Sol. Energy* **191**, 672–685 (2019).
38. Kaaya, I., Ascencio-Vásquez, J., Weiss, K.-A. & Topič, M. Assessment of uncertainties and variations in PV modules degradation rates and lifetime predictions using physical models. *Sol. Energy* **218**, 354–367 (2021).
39. Mirlletz, H., Oviatt, S., Sridhar, S. & Barnes, T. M. Prioritizing circular economy strategies for sustainable PV deployment at the TW scale. *EPJ Photovolt.* **15**, 18 (2024).
40. Peters, I. M., Hauch, J. & Brabec, C. Cradle-to-cradle recycling in terawatt photovoltaics: a vision of perpetual use. *Joule* **8**, 899–912 (2024).
41. Malik, M. M. *Extraction of Solar Cell Data from PDF Datasheets*. MSc thesis, Univ. Freiburg (2023); https://ad-publications.cs.uni-freiburg.de/theses/Master_Moez_Malik_2023.pdf
42. Malik, M. M. Lightning-table. *GitHub* <https://github.com/moezm Malik/Lightning-Table> (2025).
43. Pieters, M. PyMuPDF. *GitHub* <https://github.com/pymupdf/PyMuPDF> (2025).
44. Paszke, A. et al. PyTorch: an imperative style, high-performance deep learning library. Preprint at <https://arxiv.org/abs/1912.01703> (2019).
45. Ren, S., He, K., Girshick, R. B. & Sun, J. Faster R-CNN: towards real-time object detection with region proposal networks. Preprint at <https://arxiv.org/abs/1506.01497> (2015).
46. Aristarán, M., Tigas, M., Merrill, J.B. & Das, J. Tabula: extract tables from PDFs. *GitHub* <https://github.com/tabulapdf/tabula> (2025).
47. Mehta, V. Camelot: PDF Table Extraction for Humans v1.0.9 (Camelot Developers, 2025); <https://camelot-py.readthedocs.io/en/master/>
48. Mehta, V. Camelot v1.0.9. *GitHub* <https://github.com/camelot-dev/camelot> (2025).
49. Smith, R. An overview of the Tesseract OCR engine. In *Proc. Ninth International Conference on Document Analysis and Recognition (ICDAR 2007)* (IEEE, 2007); <https://ieeexplore.ieee.org/document/4376991>
50. Salton, G. & Buckley, C. Term-weighting approaches in automatic text retrieval. *Inf. Process. Manag.* **24**, 513–523 (1988).
51. McCallum, A. & Nigam, K. A Comparison of event models for naive Bayes text classification. In *AAAI-98 Workshop on Learning for Text Categorization* (Association for the Advancement of Artificial Intelligence, 1998); <https://aaai.org/papers/041-ws98-05-007/>
52. Pedregosa, F. et al. Scikit-learn: machine learning in Python. *J. Mach. Learn. Res.* **12**, 2825–2830 (2011).
53. Electricity generation by source, *Our World in Data*, <https://ourworldindata.org/grapher/electricity-prod-source-stacked> (2024).

Acknowledgements

This Perspective was informed by the 4th Terawatt Workshop that was convened in Pacific Grove, California, USA on 6–8 June 2024. This work was authored in part by the National Renewable Energy Laboratory for the US Department of Energy (DOE) under contract no. DE-AC36-08GO28308. The views expressed in the article do not necessarily represent the views of the DOE or the US Government. The US Government retains and the publisher, by accepting the article for publication, acknowledges that the US Government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this work, or allow others to do so, for US Government purposes. The views expressed are also based on the current information available to the authors and may not in any circumstances be regarded as stating an official or policy position of the European Commission.

Competing interests

We declare employment-based competing interests for the following: A.K. is the President of AGC Business Development Americas; AGC is a glass, materials and chemical manufacturer. C.C. is the Chief Scientist at Oxford PV, a perovskite PV solar technology company. Y.C. is Vice President of Technology for Trina Solar, a PV and energy company. A.F. is the Chief Technology Officer of SOLARWATT, a full-service provider of integrated solar systems. S.G. is Vice President of Renewable Energies and Strategic Partnerships for VON ARDENNE GmbH, a company that manufactures vacuum coating systems and processes for industrial applications. M.G. is the Chief Technology Officer at First Solar, a producer of thin-film PV systems. E.H. is a senior consultant at Swiss RE, a commercial insurance provider. B.H. is the Head of Advanced Research at First Solar. J.J. is CEO and co-founder of Swift Solar, a US company focused on perovskite tandem solar technology. S.J. is R&D manager of the North American Thin Film Technology Team at NSG Group, a manufacturer of glass and glazing products. I.K. is Director and Principal Analyst at RTS Corporation, which is a PV industry and marketing consulting company. M. Kulkarni is Chairman and CEO of Renaissance Solar and Electronic Materials, a solar crystal growth and wafer manufacturing company. D. Merfeld is the Global Chief Technology Officer for Hanwha Qcells, a company focused on energy solutions from solar cells to power plants. N.K. is an employee of the Sustainable Management Initiatives Group at AGC. S.N. is with JERA Americas, an energy infrastructure company focused on low-carbon systems. J.P. is Vice President of Business Development at NexWafe, a company focused on next-generation PV wafer production. G.R. is co-founder and technical director at OFGEN, a distributed renewable energy company in Africa. D.R. is VP Technology Strategy of Moxeon Solar Technologies, a business focused on high-efficiency solar panels. R. Sinton is President and Senior Scientist of Sinton Instruments

with an equity stake. Sinton Instruments manufactures test and measurement equipment for R&D and manufacturing in the PV industry. K. Soni is the Technology Assessment Director, Exploratory Markets and Technologies at Corning, Inc., which is a materials and manufacturing company focused on glass and ceramics. D.S. is the CEO of NexWafe. K.W. was formerly a Project Manager at bifa Umweltinstitut GmbH, which is an environmental research, development and consulting institute.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41560-025-01929-z>.

Correspondence should be addressed to Kirstin Alberi.

Peer review information *Nature Energy* thanks Martin Green and Eric O'Shaughnessy for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

This is a U.S. Government work and not under copyright protection in the US; foreign copyright protection may apply 2025

Kirstin Alberi^{1,2}✉, I. Marius Peters³, Pierre Verlinden⁴, Simon Philipps⁵, Akio Koike⁶, Teresa Barnes¹, Joe Berry^{1,2,7}, Mariana Bertoni⁸, Christian Breyer⁹, Laurie Burnham¹⁰, Chris Case¹¹, Yifeng Chen¹², Stefaan De Wolf¹³, Renate Egan¹⁴, Armin Froitzheim¹⁵, Sebastian Gatz¹⁶, Markus Gloeckler¹⁷, Jan Christoph Goldschmidt¹⁸, Ivan Gordon^{19,20,21}, Nancy M. Haegel¹, Martin Hermle⁵, Christiana Honsberg⁸, Edward Hsi²², Bill Huber¹⁷, Shogo Ishizuka²³, Arnulf Jäger-Waldau²⁴, Joel Jean²⁵, Jessica Yajie Jiang¹⁴, Shannon Jurca²⁶, Izumi Kaizuka²⁷, Richard R. King¹⁸, Keiichi Komoto²⁸, Michio Kondo^{23,29}, Milind Kulkarni³⁰, Sarah Kurtz³¹, Daniel Macdonald³², Danielle Merfeld³³, Naoya Kobayashi⁶, Shigeru Niki³⁴, Andreas Obst³⁵, Takashi Oozeki²³, Ulrich W. Paetzold^{36,37}, Jonathan Pickering³⁸, Ralf Preu⁵, Samantha B. Reese¹, Christian Reichel⁵, Thomas Reindl³⁹, Ingrid Repins¹, Geoffrey Ronoh⁴⁰, Doug Rose⁴¹, Keiichiro Sakurai²³, Rutger Schlatmann⁴², Abdelilah Slaoui⁴³, Ron Sinton⁴⁴, Kamal Soni⁴⁵, Billy J. Stanbery⁴⁶, Davor Sutija³⁸, Marko Topić⁴⁷, Yuzuru Ueda⁴⁸, Juzer Vasi⁴⁹, Karsten Wambach⁵⁰, Emily Warren¹, Eicke Weber⁵¹, Masafumi Yamaguchi⁵² & Andreas W. Bett^{5,53}

¹National Renewable Energy Laboratory, Golden, CO, USA. ²Renewable and Sustainable Energy Institute, University of Colorado Boulder, Boulder, CO, USA. ³IMD-3: Photovoltaics, Forschungszentrum Jülich GmbH, Jülich, Germany. ⁴Yangtze Institute for Solar Technology, Jiangyin, China. ⁵Fraunhofer Institute for Solar Energy Systems ISE, Freiburg, Germany. ⁶AGC Inc., Tokyo, Japan. ⁷University of Colorado Boulder, Boulder, CO, USA. ⁸Arizona State University, Tempe, AZ, USA. ⁹School of Energy Systems, LUT University, Lappeenranta, Finland. ¹⁰Sandia National Laboratory, Albuquerque, NM, USA. ¹¹Oxford Photovoltaics Ltd, Yarnton, UK. ¹²Trina Solar, State Key Laboratory of PV Science and Technology, Changzhou, China. ¹³Physical Sciences and Engineering Division, KAUST Solar Center, King Abdullah University of Science and Technology (KAUST), Thuwal, Kingdom of Saudi Arabia. ¹⁴School of Photovoltaic and Renewable Energy Engineering, The University of New South Wales, Sydney, New South Wales, Australia. ¹⁵AMFutureTech, Schwaikheim, Germany. ¹⁶VON ARDENNE GmbH, Dresden, Germany. ¹⁷First Solar, Inc., Perrysburg, OH, USA. ¹⁸Department of Physics, University of Marburg, Marburg, Germany. ¹⁹imo-imomec, Hasselt University, Hasselt, Belgium. ²⁰Imec, Genk, Belgium. ²¹Energyville, Genk, Belgium. ²²Swiss RE, Singapore, Singapore. ²³National Institute of Advanced Industrial Science and Technology, Tokyo, Japan. ²⁴European Commission, Joint Research Center, Ispra, Italy. ²⁵Swift Solar, San Carlos, CA, USA. ²⁶NSG Group, Niigata, Japan. ²⁷RTS Corporation, Tokyo, Japan. ²⁸Mizuho Research & Technology Ltd, Tokyo, Japan. ²⁹Waseda University, Tokyo, Japan. ³⁰Renaissance Solar and Electronic Materials (RSOLEC), Chennai, India. ³¹University of California, Merced, Merced, CA, USA. ³²School of Engineering, The Australian National University, Canberra, Australian Capital Territory, Australia. ³³Hanwha Qcells, Irvine, CA, USA. ³⁴JERA Americas Inc., Houston, TX, USA. ³⁵Fraunhofer Center for Silicon Photovoltaics CSP, Halle, Germany. ³⁶Institute of Microstructure Technology, Karlsruhe Institute of Technology, Eggenstein-Leopoldshafen, Germany. ³⁷Light Technology Institute, Karlsruhe Institute of Technology, Karlsruhe, Germany. ³⁸NexWafe GmbH, Freiburg, Germany. ³⁹Solar Energy Research Institute of Singapore (SERIS), National University of Singapore (NUS), Singapore, Singapore. ⁴⁰OFGEN, Nairobi, Kenya. ⁴¹Moxeon Solar Technologies, San Jose, CA, USA. ⁴²Competence Centre Photovoltaics Berlin (PVcomB), Helmholtz-Zentrum Berlin für Materialien und Energie, Berlin, Germany. ⁴³ICube laboratory, CNRS and Université de Strasbourg,

Strasbourg, France. ⁴⁴Sinton Instruments, Boulder, CO, USA. ⁴⁵Corning Inc., Corning, NY, USA. ⁴⁶Clean Energy Institute, University of Washington, Seattle, WA, USA. ⁴⁷Faculty of Electrical Engineering, University of Ljubljana, Ljubljana, Slovenia. ⁴⁸Tokyo University of Science, Tokyo, Japan. ⁴⁹National Center for Photovoltaic Research and Education, Department of Electrical Engineering, Indian Institute of Technology, Bombay, India. ⁵⁰bifa Umweltinstitut GmbH, Augsburg, Germany. ⁵¹University of California, Berkeley, Berkeley, CA, USA. ⁵²Toyota Technical Institute, Nagoya, Japan. ⁵³Physics Institute, Albert-Ludwigs-Universität Freiburg, Freiburg, Germany. ✉e-mail: Kirstin.Alberi@nrel.gov